

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think>

**Finding reasoning paths : **

- 1 . Evaluate the anatomical location of the stab wound : in the region of the anatomical snuff box , which is a common site for vascular injuries .
- 2 . Identify the major arteries in the hand , particularly in the anatomical snuff box : palmar branches of the radial , ulnar , and median arteries .
- 3 . Determine which artery is most commonly injured in the anatomical snuff box due to its location and proximity to the wound .
- 4 . Consider the physiological consequences of arterial damage : rapid bleeding due to a low-resistance artery with abundant collateral circulation .

**Reasoning Process : **

- 1 . **Anatomical Location : ** The anatomical snuff box is a pocket formed by the scaphoid , hamate , and pisiform bones , and it is located on the radial side of the hand . It is an area rich in arteries and veins , making it a common site for vascular injuries .
- 2 . **Major Arteries in the Hand : **
 - The **palmar branches of the radial artery** are the most superficial and vulnerable to injury in the anatomical snuff box . They run in a superficial location and are more exposed to trauma .
 - The **palmar branch of the ulnar artery** runs closer to the deep fascia and is less likely to be injured unless there is significant pressure or muscle damage .
 - The **palmar branch of the median artery** is the deepest and runs closely with the thenar muscles , making it less accessible to injury in the anatomical snuff box .
- 3 . **Most Commonly Injured Artery : **
 - Given the superficial location and anatomical orientation , the **radial artery** is the most likely artery to be injured in the anatomical snuff box . This is because it is the most superficial and has a larger vascular supply , making it more prone to laceration .
- 4 . **Physiological Consequences : **
 - When the **radial artery** is damaged , there is rapid bleeding due to its low-resistance flow and abundant collateral circulation . This can lead to significant blood loss , as seen in this case of severe bleeding from the right hand .

**Conclusion : **

Damage to the **radial artery** is most likely responsible for the excessive bleeding in this case , as it is the most superficial and commonly injured artery in the anatomical snuff box .

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<answer>

Radial artery

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